

HiveForce Labs

THREAT ADVISORY



VULNERABILITY REPORT

**Apple addresses macOS Dirty Cow,
Achilles, and other flaws**

Date of Publication

December 20, 2022

Admiralty Code

A1

TA Number

TA2022307

Summary

First Seen: December 2022

Affected Product: macOS Monterey

Impact: results in the complete compromise of the vulnerable system due to code execution.

CVEs

CVE	NAME	PATCH
CVE-2022-42854	Bluetooth Kernel Memory Disclosure	
CVE-2022-42821	BOM Bypass Gatekeeper checks	
CVE-2022-32942	DriverKit arbitrary code execution	
CVE-2022-42861	File System sandbox breakout	
CVE-2022-42864	IOHIDFamily arbitrary code execution	
CVE-2022-46689	Kernel arbitrary code execution	
CVE-2022-42845	Kernel arbitrary code execution	
CVE-2022-42842	Kernel code execution	

CVE	NAME	PATCH
CVE-2022-40303	Libxml2 arbitrary code execution	
CVE-2022-40304	Libxml2 arbitrary code execution	
CVE-2022-42840	Ppp arbitrary code execution	
CVE-2022-42855	Preferences logic issue	
CVE-2022-42841	Xar arbitrary code execution	

Vulnerability Details

#1

Apple addressed multiple vulnerabilities in macOS Monterey. These vulnerabilities affect different functionalities such as Bluetooth, BOM, DriverKit, File System, IOHIDFamily, Kernel, and Preferences. Some of these vulnerabilities could be exploited by an attacker to execute arbitrary code.

#2

CVE-2022-46689 has been called the macOS equivalent of Dirty Cow, which is a vulnerability allowing a locally authenticated attacker to gain elevated privileges on the system, caused by a race condition in the macOS kernel. It is possible for an authenticated attacker to exploit this vulnerability by sending a specially-crafted request.

#3

Achilles is a security flaw identified by CVE-2022-42821. Malware and other threats might be able to leverage this Gatekeeper bypass vulnerability to gain initial access to macOS and increase the success rate of attacks and campaigns.

Vulnerability

CVE ID	AFFECTED PRODUCTS	AFFECTED CPE	CWE ID
CVE-2022-42854	macOS: 12.0 21A344 - 12.6.1 21G217	cpe:2.3:o:apple:macos*: *.***.*.*.*.*	CWE-125
CVE-2022-42821			CWE-254
CVE-2022-32942			CWE-119
CVE-2022-42861			CWE-269
CVE-2022-42864			CWE-362
CVE-2022-46689			CWE-362
CVE-2022-42845			CWE-119
CVE-2022-42842			CWE-119
CVE-2022-40303			CWE-190
CVE-2022-40304			CWE-399
CVE-2022-42840			CWE-119
CVE-2022-42855			CWE-371
CVE-2022-42841			CWE-843

Recommendations



Security Leaders

Asset and vulnerability management solutions should be implemented to ensure that all internet-accessible devices are secure, patched, updated, hardened, and monitored. Integrate and communicate all lessons learned.



Security Engineers

- **Uni5 Users:** This is an actionable threat advisory in HivePro Uni5. Prioritize the vulnerability, identify the impacted assets, and patch them through your Command Center. Test your controls with Uni5's Breach & Attack Simulation.
- **All Engineers:** Refer to and action upon the 'Potential MITRE ATT&CK TTPs' & 'Patch Link' on the following pages.

Potential MITRE ATT&CK TTPs

<u>TA0042</u> Resource Development	<u>TA0001</u> Initial Access	<u>TA0004</u> Privilege Escalation	<u>TA0005</u> Defense Evasion
<u>T1548</u> Abuse Elevation Control Mechanism	<u>T1190</u> Exploit Public-Facing Application	<u>T1588</u> Obtain Capabilities	<u>T1588.006</u> Vulnerabilities
<u>T1553</u> Subvert Trust Controls	<u>T1553.001</u> Gatekeeper Bypass		

Patch Details

Upgrade to macOS Monterey 12.6.2

References

<https://support.apple.com/en-ae/HT213533>

<https://worthdoingbadly.com/macdirtycow/>

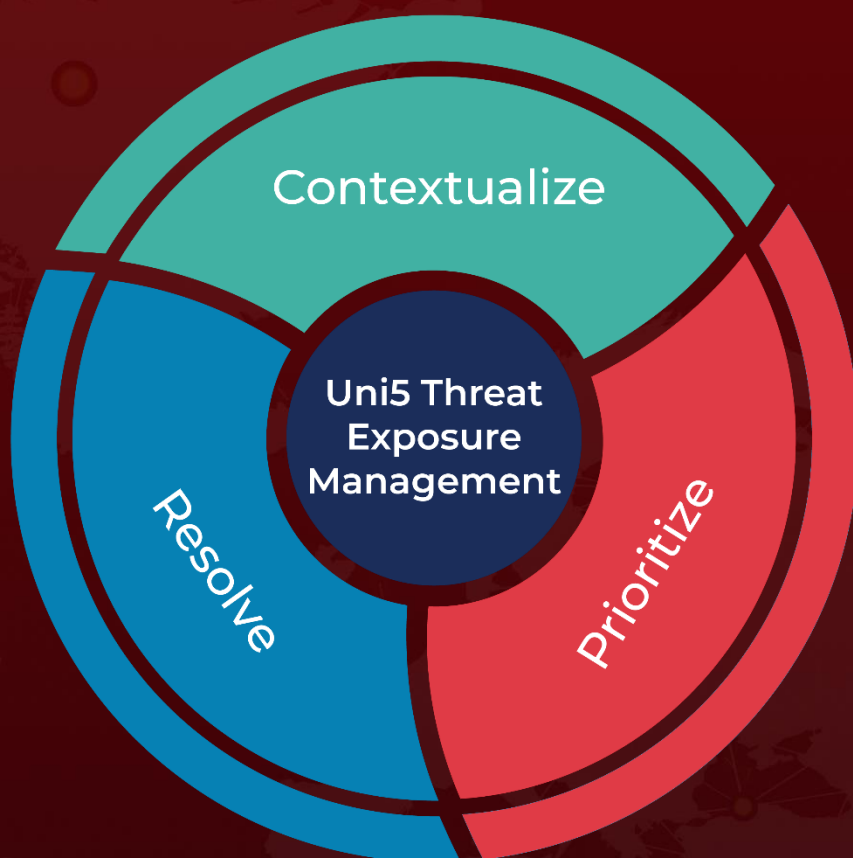
<https://github.com/zhuowei/MacDirtyCowDemo>

<https://www.microsoft.com/en-us/security/blog/2022/12/19/gatekeepers-achilles-heel-unearthing-a-macos-vulnerability/>

What Next?

At Hive Pro, it is our mission to detect the most likely threats to your organization and to help you prevent them from happening.

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